

AvatarForge : A Real-Time and Node-Based Body Editing System

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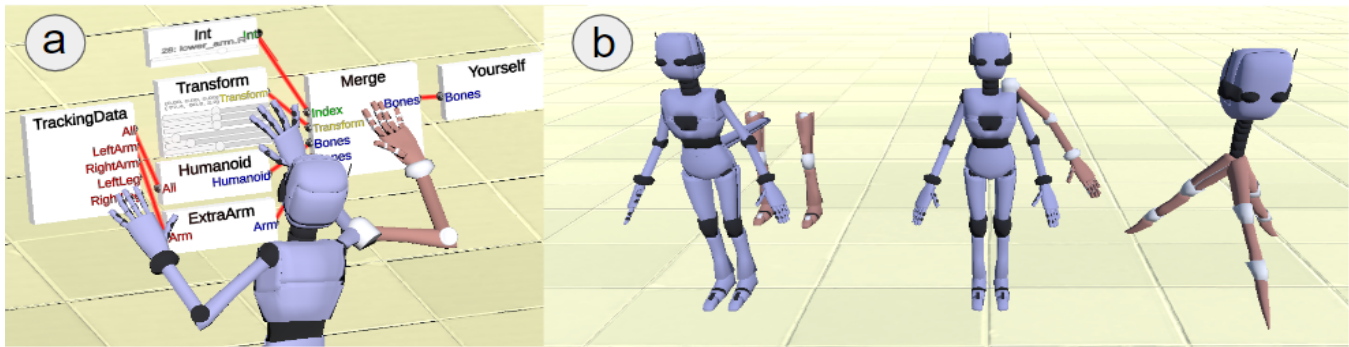


Figure 1: (a) A player designing their own avatar in real-time in a virtual environment. (b) Avatars designed with AvatarForge.

ABSTRACT

Designing non-humanoid avatars with different body structures from humans so that humans can intuitively manipulate them requires the design of body structure and motion mapping, which is complex, takes time and requires special design skills. To address this problem, we propose AvatarForge which allows users to design non-humanoid avatars by editing body structure and motion mapping in real-time. In this system, users interact with a node-based interface in a virtual environment to edit their own bodies while visualizing the changes. This system aims to reduce the difficulty of designing non-humanoid avatars, expedite the prototyping of new bodies, and enable personalized avatars that match individual preferences and physical abilities. It contributes to the advancement of non-humanoid body utilization in various fields such as CG animation, inclusive design, and human augmentation.

KEYWORDS

Virtual Reality, Human Augmentation, Body Scheme Alternation

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1 INTRODUCTION

By utilizing non-humanoid avatars with a different structure from the typical humanoid body, humans can surpass their innate physical limitations and acquire new bodily experiences, expressive capabilities, and functional skills. For example, Human tails [Step-toe et al. 2013] shows that adding a tail to a humanoid avatar and properly manipulating it improves human motor performance. In VR games and social VR services, people are using non-humanoid avatars, such as animal avatars, to perform various tasks and self-expressions using the unique characteristics of these bodies.

There are two critical parameters specific to the design of non-humanoid avatars: structure and motion mapping. The avatar's structure is represented by a series of bones and has a significant impact on the experiences and functionalities it can provide to the user. On the other hand, motion mapping, which refers to how the user's body movements map to the avatar's movements, influences the intuitiveness and accuracy of the avatar's control.

There are several existing studies on methods of creating motion mapping for non-humanoid avatars. In the field of CG animation, techniques have been proposed to animate objects, such as lamps [Yamane et al. 2010], by matching their poses to human poses, thereby synchronizing their movements with human actions. Furthermore, HandAvatar [Jiang et al. 2023] has proposed an algorithm to generate optimal motion mappings from hand to non-humanoid avatars based on the result of user studies.

However, it is difficult to edit the mappings generated by these methods and to optimize them for each individual user. To enable users, including those with physical disabilities, to design task-specific and personalized non-humanoid avatars, we propose AvatarForge, which enables users to design non-humanoid avatars by interactively reconfiguring their own body structure and motion mapping in real time in VR.

2 SYSTEM DESIGN

In AvatarForge, users can design various avatars with node-based compositing tool in the virtual environment. The designed avatars function as their own bodies, allowing users to constantly preview the real-time results of their node-based edits. Nodes can be divided into the following five groups:

2.1 Structure Node Group

Each node in this group has a corresponding structure specific to that node. Two foundation nodes are provided as a basis for augmenting the body structure: the "Humanoid" and the "Non-humanoid spine". The former retains the structure of a typical humanoid avatar and is therefore suitable for creating extended-humanoid avatars that have new body parts added to the humanoid base, such as the avatar with an additional arm shown in the example in Figure 1.b (left and center). The latter, on the other hand, retains only the bone structure of a humanoid avatar from head to hips, making it suitable for creating avatars that break away from the constraints of the regular human body, such as the three-legged alien depicted in Figure 1.b (right), or animal-like avatars. In addition to these basic nodes, the structures of different body parts such as arms, legs, wings, tails, and tentacles are provided as nodes and can be used in combination with the basic nodes.

2.2 Motion Node Group

Each node in this group has a corresponding motion. For example, the "TrackingData" node contains the user's own body motion. Users can add more tracking systems in addition to the HMD and hand controllers if they need more degrees of freedom to control the new bodies. There are also nodes that hold motions such as looping animations. Connecting a *Motion* node to a *structure* node animates the structure associated with that *structure* node.

2.3 Modification Node Group

This node group allows operations on the structure of *structure* nodes. For example, a "Merge" node can combine two structures to create a new structure, and an "IK" node can set up inverse kinematics to simplify the structure. Using *Modification* nodes, complex body structures can be easily designed. In addition, each *Modification* node also has a structure, just like *Structure* nodes, so users can re-edit with other *Modification* nodes. This allows users to easily create unique and complex avatar designs.

2.4 Utility Node Group

Each node in this group stores various numeric parameters. For example, the "Int" node can be used to provide numeric input to the *Modification* node. For instance, it can be used to specify the bone hierarchy for merging two structures using the "Merge" node.

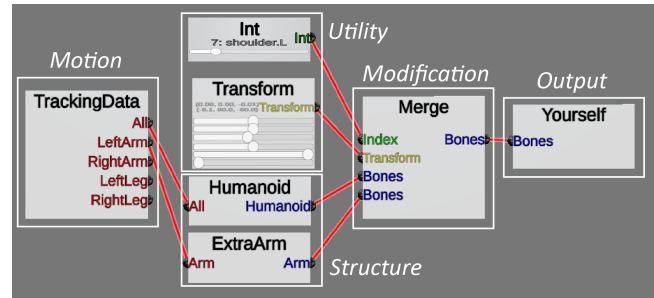


Figure 2: States of the node of the avatar with an extra arm(Fig 1.b center).

2.5 Output Node Group

This node performs as a terminal node. The structure of a *Structure* node or *Modification* node connected to this node serves as the user's body that can be utilized.

3 DEMONSTRATION AT SIGGRAPH ASIA

At SIGGRAPH Asia'23, visitors will be able to create avatars using our system. They can experience the simplicity of the creation process and test the performance of their avatars in various scenes, such as a game scene with floating targets and a scene where multiple users interact using non-humanoid bodies.

4 DISCUSSION AND FUTURE WORK

This system aims to reduce the design difficulty for non-humanoid avatars, expedite prototyping, and enable personalized avatars based on individual preferences and physical abilities. As a result, the system is expected to promote the use of non-humanoid bodies in various fields such as CG animation, inclusive design, and human augmentation. In addition, the real-time body editor systems can be integrated into VR games, making a new gaming experience.

In the future, we will conduct workshops where the general public can create avatars, to evaluate the user experience and analyze the designed avatars. Based on the analysis results, we plan to add functions such as suggesting appropriate body structure and motion mapping for desired abilities, enabling collaborative avatar design between the users and the system.

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